

Justine Abanilla

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EDUCATION

Colegio De Montalban

Bachelor of Science in Computer Engineering

Rodriguez Rizal

2023 – Present

TECHNICAL SKILLS

- **Languages:** Python (Pandas, FastAPI, BeautifulSoup/Scrapy), SQL (PostgreSQL)
- **Data Engineering:** PostgreSQL, Pandas, dbt, Snowflake, Apache Airflow, Docker, Metabase
- **Cloud:** AWS (S3), Azure (Blob Storage)
- **Tools & Platforms:** Git, GitHub, VS Code, Postman

CERTIFICATIONS

- **DataCamp** — Data Engineering Professional | Data Engineering Associate | SQL Associate | Python Data Associate | Github Copilot Fundamentals (GH-300)
- **Claude** — Claude 101

EXPERIENCE

FlyRank

Remote

Machine Learning Engineer Intern (Cohort-Based, Capstone Program)

June 2026 – Aug 2026

- Selected for a cohort-based ML internship culminating in a published capstone: a content-refresh opportunity ranking system for FlyRank's editorial review pipeline, built end-to-end from an anonymized page-metrics dataset through model validation and a reproducible codebase.
- Engineered a leakage-safe feature set spanning visibility (impressions, average position), freshness (days since update, content age), and engagement (CTR, scroll rate, engagement rate) signals, then trained and compared decision tree, logistic regression, and random forest classifiers against a deterministic rule-based baseline.
- Achieved a Precision@50 of 0.740 and ROC AUC of 0.750 with the random forest model, a substantial improvement over the baseline's 0.240 and 0.627 respectively, using a client-aware holdout split so training and test clients never overlapped.
- Ran leakage audits to exclude label-derived and future-performance features, keeping the model restricted to pre-decision signals only.
- Translated model output into a ranked action playbook (monitor / refresh / refresh-and-review-CTR / expand-and-refresh) with reason-code explainability, designed as a human-in-the-loop reviewer aid rather than an automated publishing engine.
- Documented the full pipeline for reproducibility across structured notebooks (feature/leakage checks, baseline scoring, modeling, validation audit, action playbook) and a single run script, with results and charts published as part of the capstone report.

PROJECTS

- **SAIS – Soil Accommodation Intelligence System** (Thesis Project): An MLOps pipeline for real-time soil monitoring that predicts soil moisture (SGD Regressor) and classifies soil fertility (Random Forest), with Airflow scheduling live inference and periodic online retraining on incoming sensor data. Includes a 53-test suite covering the full ingestion-to-retraining cycle.
- **Content Refresh Opportunity Ranking (FlyRank Capstone):** A random forest model that ranks web pages by refresh priority using traffic and engagement signals, beating a rule-based baseline (0.740 vs. 0.240 Precision@50).
- **PH Weather Pulse** : Hourly pipeline that pulls live weather data for 5 PH cities via API, stores it in PostgreSQL, transforms it with dbt, and visualizes it in a Metabase dashboard — orchestrated by Airflow and containerized with Docker; Snowflake sync in progress.